Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

Code :-

```
# COMPLETE CAPSTONE PROJECT
# Programming Error Analysis + Machine Learning Model

# Load Required Packages Safely
required_packages <- c("ggplot2","caret","randomForest","e1071","nnet")

for(pkg in required_packages){
  if(!require(pkg, character.only = TRUE)){
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}

set.seed(123)

# PART 1: DESCRIPTIVE STATISTICAL ANALYSIS
Error_Type <- c("Syntax Errors", "Semantic Errors", "Logical Errors")
Frequency <- c(30, 28, 42)

data_desc <- data.frame(Error_Type, Frequency)
data_desc$Percentage <- (data_desc$Frequency / sum(data_desc$Frequency)) * 100

cat("=== DESCRIPTIVE DATA TABLE ===\n")
print(data_desc)

cat("\nTotal Errors:", sum(data_desc$Frequency))
cat("\nMean Errors:", mean(data_desc$Frequency))
cat("\nStandard Deviation:", sd(data_desc$Frequency), "\n")

# Bar Plot
barplot(data_desc$Frequency,
        names.arg = data_desc$Error_Type,
        col = c("steelblue","orange","forestgreen"),
        main = "Programming Error Distribution",
        ylab = "Frequency")

# Pie Chart
pie(data_desc$Frequency,
```

Name :- Vinod Manohar Mali
Roll No :- S092

```
    labels = paste(data_desc$Error_Type,
                   round(data_desc$Percentage,2), "%"),
    col = c("steelblue","orange","forestgreen"),
    main = "Programming Error Percentage")

# Chi-Square Test
cat("\n=== Chi-Square Test ===\n")
print(chisq.test(data_desc$Frequency))

# ggplot Visualization
ggplot(data_desc, aes(x=Error_Type, y=Frequency, fill=Error_Type)) +
  geom_bar(stat="identity") +
  theme_minimal() +
  ggtitle("Programming Error Distribution (ggplot)")

# Export CSV
write.csv(data_desc,
          "Programming_Error_Descriptive.csv",
          row.names = FALSE)

# PART 2: MACHINE LEARNING PREDICTIVE MODEL
n <- 300

Practice_Hours <- rnorm(n, mean=8, sd=3)
Compiler_Exposure <- rnorm(n, mean=15, sd=5)
Concept_Score <- rnorm(n, mean=65, sd=10)
Debug_Time <- rnorm(n, mean=25, sd=8)
Experience_Level <- sample(c("Beginner","Intermediate"),
                           n, replace=TRUE)

# Target variable creation
Error_Type <- ifelse(Concept_Score < 60 & Debug_Time > 25, "Logical",
                     ifelse(Compiler_Exposure < 12, "Syntax", "Semantic"))

data_ml <- data.frame(Practice_Hours,
                      Compiler_Exposure,
                      Concept_Score,
                      Debug_Time,
                      Experience_Level,
```

```
                    Error_Type)

# Convert to factor
data_ml$Error_Type <- as.factor(data_ml$Error_Type)
data_ml$Experience_Level <- as.factor(data_ml$Experience_Level)

# Split Data
trainIndex <- createDataPartition(data_ml$Error_Type,
                                  p=0.8,
                                  list=FALSE)

trainData <- data_ml[trainIndex,]
testData  <- data_ml[-trainIndex,]

# Ensure factor consistency
trainData$Error_Type <- as.factor(trainData$Error_Type)
testData$Error_Type  <- as.factor(testData$Error_Type)


# Random Forest Model
rf_model <- randomForest(Error_Type ~ .,
                         data=trainData,
                         ntree=200)

rf_pred <- predict(rf_model, testData)

cat("\n=== Random Forest Results ===\n")
print(confusionMatrix(rf_pred, testData$Error_Type))

# Feature Importance
print(importance(rf_model))
varImpPlot(rf_model)

# Support Vector Machine
svm_model <- svm(Error_Type ~ ., data=trainData)

svm_pred <- predict(svm_model, testData)

cat("\n=== SVM Results ===\n")
```

```
print(confusionMatrix(svm_pred, testData$Error_Type))

# Multinomial Logistic Regression
multi_model <- multinom(Error_Type ~ ., data=trainData, trace=FALSE)

multi_pred <- predict(multi_model, testData)

cat("\n=== Logistic Regression Results ===\n")
print(confusionMatrix(multi_pred, testData$Error_Type))

# Accuracy Calculation
accuracy <- sum(rf_pred == testData$Error_Type) /
  nrow(testData)

cat("\nFinal Random Forest Accuracy:", accuracy)

# Export ML dataset
write.csv(data_ml,
          "Programming_Error_ML_Dataset.csv",
          row.names = FALSE)
```

OutPut :-

```
    element

Loading required package: nnet

> set.seed(123)
> # PART 1: DESCRIPTIVE STATISTICAL ANALYSIS
> Error_Type <- c("Syntax Errors", "Semantic Errors", "Logical Errors")
> Frequency <- c(30, 28, 42)
> data_desc <- data.frame(Error_Type, Frequency)
> data_desc$Percentage <- (data_desc$Frequency / sum(data_desc$Frequency)) * 100
> cat("=== DESCRIPTIVE DATA TABLE ===\n")
=== DESCRIPTIVE DATA TABLE ===
> print(data_desc)
       Error_Type Frequency Percentage
1   Syntax Errors        30         30
2 Semantic Errors        28         28
3  Logical Errors        42         42
> cat("\nTotal Errors:", sum(data_desc$Frequency))

Total Errors: 100
> cat("\nMean Errors:", mean(data_desc$Frequency))

Mean Errors: 33.33333
> cat("\nStandard Deviation:", sd(data_desc$Frequency), "\n")

Standard Deviation: 7.571878
```

```
> # Load Required Packages Safely
> required_packages <- c("ggplot2","caret","randomForest","e1071","nnet")
> for(pkg in required_packages){
+   if(!require(pkg, character.only = TRUE)){
+     install.packages(pkg)
+     library(pkg, character.only = TRUE)
+   }
+ }

Loading required package: ggplot2
Loading required package: caret
Loading required package: lattice
Loading required package: randomForest
randomForest 4.7-1.2
Type rfNews() to see new features/changes/bug fixes.

Attaching package: 'randomForest'

The following object is masked from 'package:ggplot2':

    margin

Loading required package: e1071

Attaching package: 'e1071'

The following object is masked from 'package:ggplot2':
```

Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

```
> # Bar Plot
> barplot(data_desc$Frequency,
+         names.arg = data_desc$Error_Type,
+         col = c("steelblue","orange","forestgreen"),
+         main = "Programming Error Distribution",
+         ylab = "Frequency")
> # Pie Chart
> pie(data_desc$Frequency,
+     labels = paste(data_desc$Error_Type,
+                    round(data_desc$Percentage,2), "%"),
+     col = c("steelblue","orange","forestgreen"),
+     main = "Programming Error Percentage")
> # Chi-Square Test
> cat("\n=== Chi-Square Test ===\n")

=== Chi-Square Test ===
> print(chisq.test(data_desc$Frequency))

        Chi-squared test for given probabilities
```

```
        Chi-squared test for given probabilities

data:  data_desc$Frequency
X-squared = 3.44, df = 2, p-value = 0.1791

> # ggplot Visualization
> ggplot(data_desc, aes(x=Error_Type, y=Frequency, fill=Error_Type)) +
+   geom_bar(stat="identity") +
+   theme_minimal() +
+   ggtitle("Programming Error Distribution (ggplot)")
> # Export CSV
> write.csv(data_desc,
+           "Programming_Error_Descriptive.csv",
+           row.names = FALSE)
> # PART 2: MACHINE LEARNING PREDICTIVE MODEL
> n <- 300
> Practice_Hours <- rnorm(n, mean=8, sd=3)
> Compiler_Exposure <- rnorm(n, mean=15, sd=5)
> Concept_Score <- rnorm(n, mean=65, sd=10)
> Debug_Time <- rnorm(n, mean=25, sd=8)
> Experience_Level <- sample(c("Beginner","Intermediate"),
+                            n, replace=TRUE)
> # Target variable creation
> Error_Type <- ifelse(Concept_Score < 60 & Debug_Time > 25, "Logical",
+                      ifelse(Compiler_Exposure < 12, "Syntax", "Semantic"))
> data_ml <- data.frame(Practice_Hours,
+                       Compiler_Exposure,
```

Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

```
=== Random Forest Results ===
> print(confusionMatrix(rf_pred, testData$Error_Type))
Confusion Matrix and Statistics

          Reference
Prediction Logical Semantic Syntax
  Logical        6        0      0
  Semantic       0       38      0
  Syntax         1        0     14

Overall Statistics

               Accuracy : 0.9831
                 95% CI : (0.9091, 0.9996)
    No Information Rate : 0.6441
    P-Value [Acc > NIR] : 1.792e-10

                  Kappa : 0.9669

 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: Logical Class: Semantic Class: Syntax
Sensitivity                 0.8571          1.0000        1.0000
Specificity                 1.0000          1.0000        0.9778
```

```
Specificity                 1.0000          1.0000        0.9778
Pos Pred Value              1.0000          1.0000        0.9333
Neg Pred Value              0.9811          1.0000        1.0000
Prevalence                  0.1186          0.6441        0.2373
Detection Rate              0.1017          0.6441        0.2373
Detection Prevalence        0.1017          0.6441        0.2542
Balanced Accuracy           0.9286          1.0000        0.9889
> # Feature Importance
> print(importance(rf_model))
                   MeanDecreaseGini
Practice_Hours           5.2787821
Compiler_Exposure       71.9920156
Concept_Score           28.1943818
Debug_Time              20.4586104
Experience_Level         0.5072059
> varImpPlot(rf_model)
> # Support Vector Machine
> svm_model <- svm(Error_Type ~ ., data=trainData)
> svm_pred <- predict(svm_model, testData)
> cat("\n=== SVM Results ===\n")

=== SVM Results ===
> print(confusionMatrix(svm_pred, testData$Error_Type))
Confusion Matrix and Statistics

          Reference
Prediction Logical Semantic Syntax
```

Name :- Vinod Manohar Mali

Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

```
          Reference
Prediction Logical Semantic Syntax
  Logical        5        0      0
  Semantic       1       37      2
  Syntax         1        1     12

Overall Statistics

               Accuracy : 0.9153
                 95% CI : (0.8132, 0.9719)
    No Information Rate : 0.6441
    P-Value [Acc > NIR] : 1.635e-06

                  Kappa : 0.8295

 Mcnemar's Test P-Value : 0.5062

Statistics by Class:

                     Class: Logical Class: Semantic Class: Syntax
Sensitivity                ‍ 0.71429          0.9737        0.8571
Specificity                1.00000          0.8571        0.9556
Pos Pred Value             1.00000          0.9250        0.8571
Neg Pred Value             0.96296          0.9474        0.9556
Prevalence                 0.11864          0.6441        0.2373
Detection Rate             0.08475          0.6271        0.2034
Detection Prevalence       0.08475          0.6780        0.2373
Balanced Accuracy          0.85714          0.9154        0.9063
```

```
> # Multinomial Logistic Regression
> multi_model <- multinom(Error_Type ~ ., data=trainData, trace=FALSE)
> multi_pred <- predict(multi_model, testData)
> cat("\n=== Logistic Regression Results ===\n")

=== Logistic Regression Results ===
> print(confusionMatrix(multi_pred, testData$Error_Type))
Confusion Matrix and Statistics

          Reference
Prediction Logical Semantic Syntax
  Logical        5        2      0
  Semantic       1       36      4
  Syntax         1        0     10

Overall Statistics

               Accuracy : 0.8644
                 95% CI : (0.7502, 0.9396)
    No Information Rate : 0.6441
    P-Value [Acc > NIR] : 0.00014

                  Kappa : 0.7256

 Mcnemar's Test P-Value : 0.14895

Statistics by Class:
```

Frequency

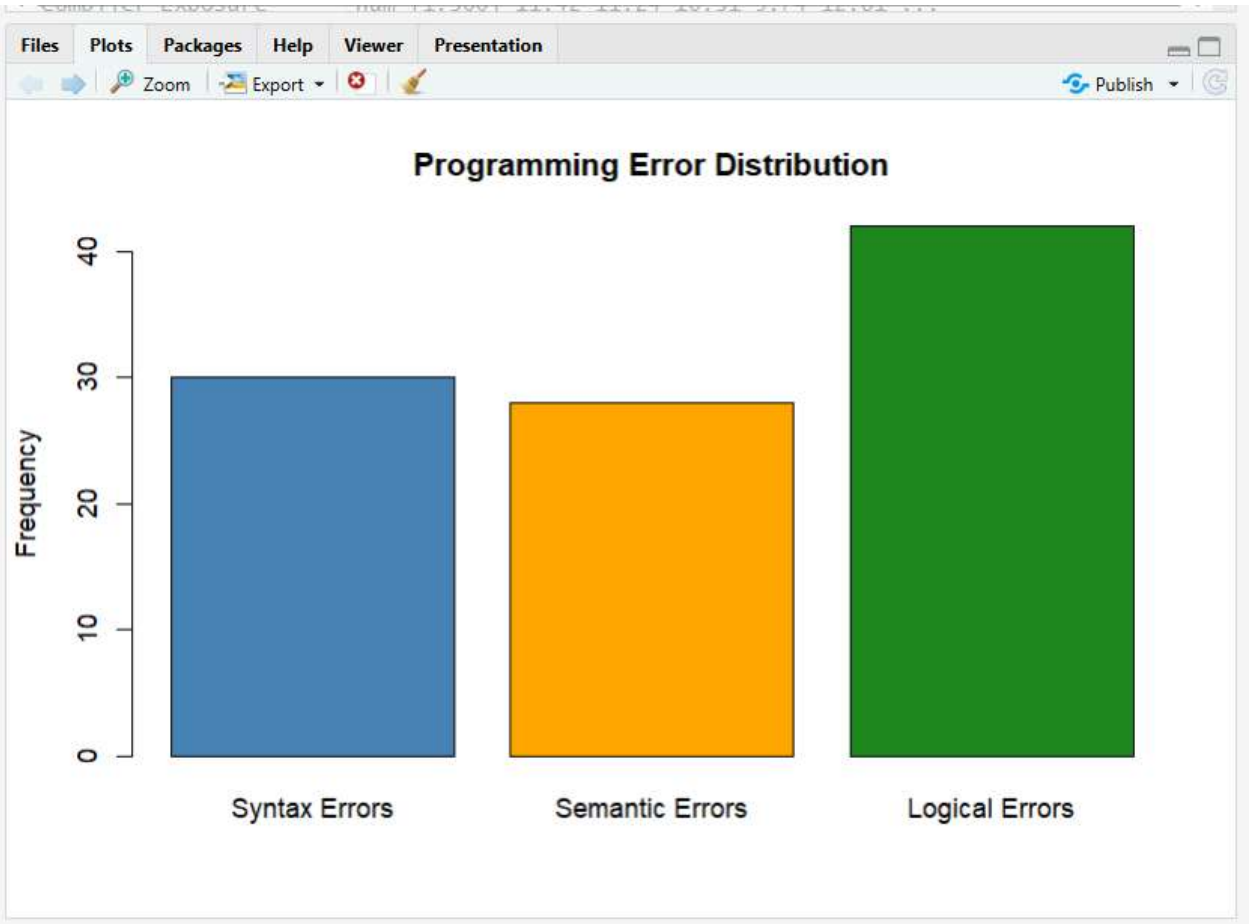
```
                     Class: Logical Class: Semantic Class: Syntax
Sensitivity                0.71429          0.9474        0.7143
Specificity                0.96154          0.7619        0.9778
Pos Pred Value             0.71429          0.8780        0.9091
Neg Pred Value             0.96154          0.8889        0.9167
Prevalence                 0.11864          0.6441        0.2373
Detection Rate             0.08475          0.6102        0.1695
Detection Prevalence       0.11864          0.6949        0.1864
Balanced Accuracy          0.83791          0.8546        0.8460
> # Accuracy Calculation
> accuracy <- sum(rf_pred == testData$Error_Type) /
+   nrow(testData)
> cat("\nFinal Random Forest Accuracy:", accuracy)

Final Random Forest Accuracy: 0.9830508
> # Export ML dataset
> write.csv(data_ml,
+           "Programming_Error_ML_Dataset.csv",
+           row.names = FALSE)
>
```

Name :- Vinod Manohar Mali

Roll No :- S092

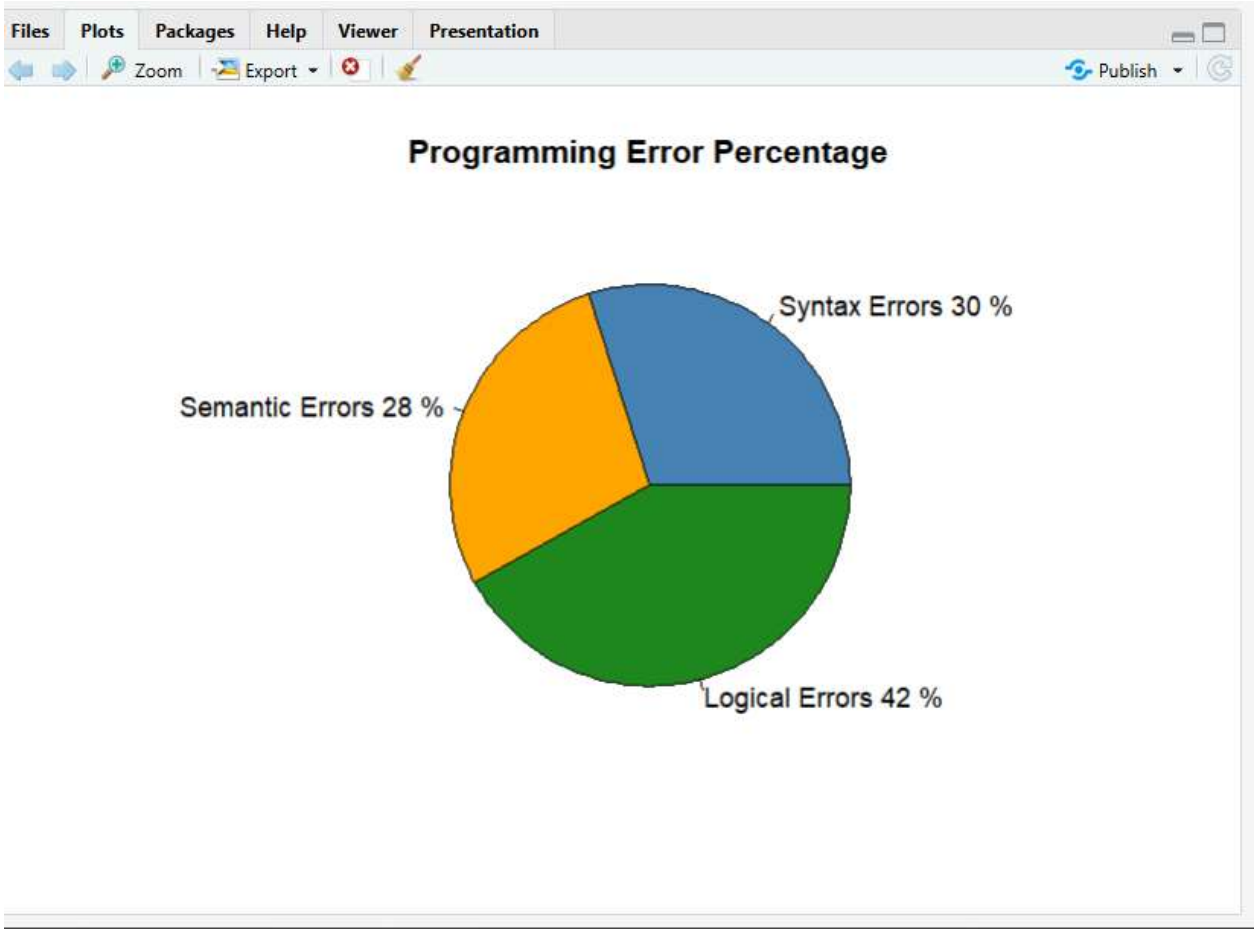
Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

Graphs :-



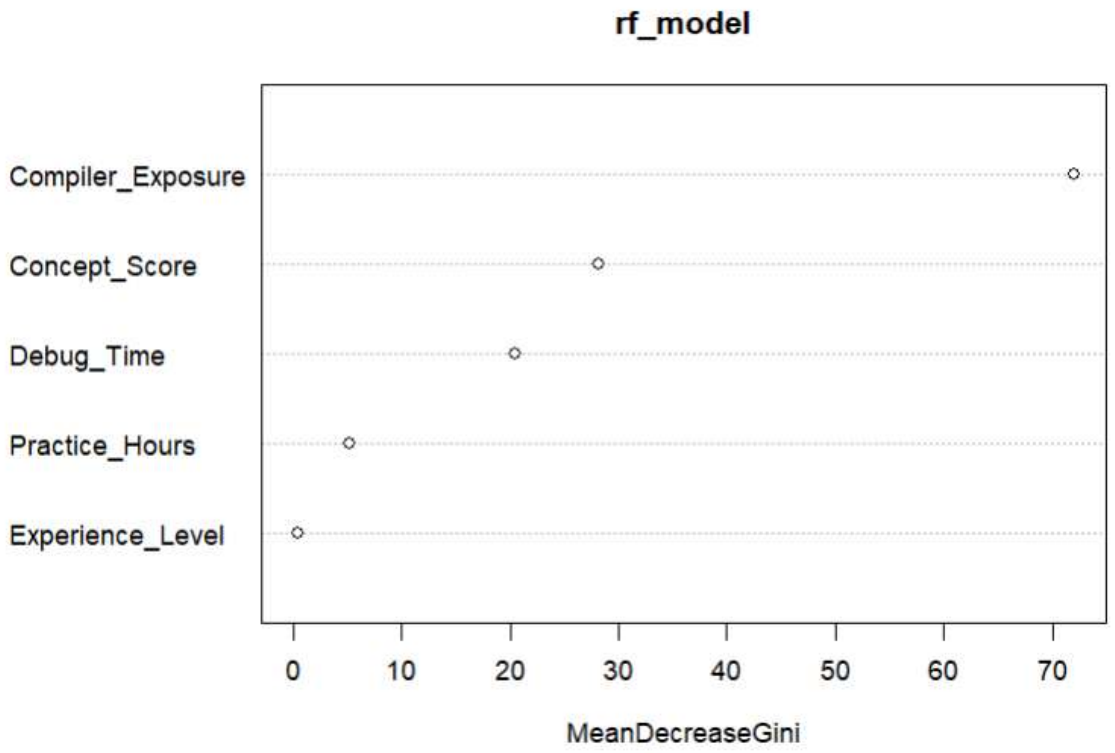
Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project



Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project



Datasets created:-
Data_desc

Filter

	Error_Type	Frequency	Percentage
1	Syntax Errors	30	30
2	Semantic Errors	28	28
3	Logical Errors	42	42

Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

Data_ml

Filter

	Practice_Hours	Compiler_Exposure	Concept_Score	Debug_Time	Experience_Level	Error_Type
1	6.318573	11.423789	75.74012	16.887087	Intermediate	Syntax
2	7.309468	11.236555	64.72653	18.669489	Beginner	Syntax
3	12.676125	10.307306	64.66670	27.396749	Beginner	Syntax
4	8.211525	9.737434	49.83932	38.112415	Beginner	Logical
5	8.387863	12.814202	72.90385	33.676936	Intermediate	Semantic
6	13.145195	16.655896	62.89266	20.003460	Beginner	Semantic
7	9.382749	4.928948	58.43257	31.607383	Intermediate	Logical
8	4.204816	16.059902	50.87974	24.611453	Beginner	Semantic
9	5.939441	21.183375	62.00238	27.410509	Beginner	Semantic
10	6.663014	25.187870	56.50939	27.082892	Beginner	Logical
11	11.672245	21.505880	61.02969	45.603598	Intermediate	Semantic

Showing 1 to 12 of 300 entries, 6 total columns

testData

Filter

	Practice_Hours	Compiler_Exposure	Concept_Score	Debug_Time	Experience_Level	Error_Type
13	9.202314	6.366348	81.87589	25.80736	Intermediate	Syntax
19	10.104068	23.422179	52.77074	28.30216	Beginner	Logical
25	6.124882	12.385438	56.24745	30.32128	Beginner	Logical
34	10.634400	22.596089	68.10230	23.22939	Beginner	Semantic
47	6.791345	15.496638	56.69188	19.69914	Beginner	Semantic
56	12.549412	16.669515	56.71826	12.02513	Intermediate	Semantic
58	9.753841	14.834819	70.38482	27.48360	Intermediate	Semantic
63	7.000378	16.368832	84.66248	26.25362	Intermediate	Semantic
75	5.935974	17.550663	65.91267	32.88847	Beginner	Semantic
87	11.290517	17.281182	62.60971	24.34330	Intermediate	Semantic
91	10.980512	11.764772	64.08566	30.40610	Beginner	Syntax

Showing 1 to 12 of 59 entries, 6 total columns

Name :- Vinod Manohar Mali
Roll No :- S092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis with SAS / SPSS /R
Capstone R Project

trainData

Filter

	Practice_Hours	Compiler_Exposure	Concept_Score	Debug_Time	Experience_Level	Error_Type
1	6.318573	11.423789	75.74012	16.887087	Intermediate	Syntax
2	7.309468	11.236555	64.72653	18.669489	Beginner	Syntax
3	12.676125	10.307306	64.66670	27.396749	Beginner	Syntax
4	8.211525	9.737434	49.83932	38.112415	Beginner	Logical
5	8.387863	12.814202	72.90385	33.676936	Intermediate	Semantic
6	13.145195	16.655896	62.89266	20.003460	Beginner	Semantic
7	9.382749	4.928948	58.43257	31.607383	Intermediate	Logical
8	4.204816	16.059902	50.87974	24.611453	Beginner	Semantic
9	5.939441	21.183375	62.00238	27.410509	Beginner	Semantic
10	6.663014	25.187870	56.50939	27.082892	Beginner	Logical
11	11.672245	21.505880	61.02969	45.603598	Intermediate	Semantic

Showing 1 to 12 of 241 entries, 6 total columns

Name :- Vinod Manohar Mali
Roll No :- S092